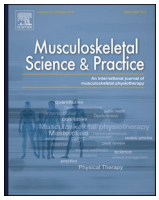
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Systematic review



Effectiveness of conservative treatment in the management of post-traumatic elbow stiffness: A systematic review

Alberto Piacenza [a,b,*], Andrea Zerilli [a,c], Ilenia Viccari [d], Greta Castelli [d]

[a] Department of Neuroscience, Rehabilitation, Ophtalmology, Genetics, Maternal and Child Health, University of Genoa, Genoa, GE, Italy
[b] Department of Territorial Care, Local Healthcare Unit 2, Savona, SV, Italy
[c] Functional Reeducation Unit, IRCCS San Martino, Largo R. Benzi, 10, 16132, Genova, Italy
[d] Department of Recovery and Functional Reeducation, La Colletta Hospital, Local Healthcare Unit 3, Arenzano, GE, Italy

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ABSTRACT

Background: Post-traumatic elbow stiffness is a common consequence following trauma or surgery, resulting in significant limb disability, with a negative impact on daily life. Although conservative treatment is the first-line approach, it is not yet known which is most suitable and effective.
Objective: To investigate the effectiveness of conservative treatments in patients with post-traumatic elbow stiffness.
Method: A protocol for this systematic review was published in PROSPERO (CRD42024517823). PRISMA standards were followed. An extensive systematic search was conducted in six databases (PubMed, CINHAL, Cochrane Library, Web of Science, Scopus, and PEDro), and the CENTRAL trial register. Two reviewers independently assessed, selected results, collected data, rated the risk of bias (RoB) of included studies with the Cochrane risk of bias tool, synthesized the available evidence, and rated it using GRADE methodology.
Results: Five studies were included in the review, although high variability in interventions and comparators precluded the synthesis of results into a meta-analysis. Large effect sizes were observed when conservative treatment was initiated immediately after immobilization, improving elbow functionality (SMD 3.07; 95%CI 1.91 to 4.23), and pain (SMD 1.83; 95%CI 0.91 to 2.76). Results indicate that Proprioceptive Neuromuscular Facilitation (SMD = −1.22; 95%CI [-1.90, −0.54]) and Graded Motor Imagery (SMD = −2.79; 95%CI [-3.59, −2.00]) were more effective than comparisons in recovering elbow functionality and pain reduction.
Conclusion: Although conservative treatment is recommended as a first-line approach, the best conservative treatment cannot be determined with certainty due to the low to very low confidence in the results.

1. Introduction

Elbow stiffness is a common consequence following trauma or surgery, with an incidence ranging from 5% to 56% (Anakwe et al., 2011; Schrumpf et al., 2013), resulting in significant limb disability, psychological stress, and a negative impact on daily life (Liu et al., 2022; Siemensma et al., 2023). The functional range of motion to perform many ADL tasks requires a degree of elbow flexion higher than 130°, such as the use of a mobile phone, computer mouse or keyboard, an elbow extension of less than 20°, as opening a door or turning a steering wheel, and more than 50° each of protonation and supination (Latz et al., 2019; Oosterwijk et al., 2018; Sardelli et al., 2011). As a result, patients' functional range should be determined individually, considering their occupational and sport's needs, hand dominance and strength, and should be considered while designing rehabilitation programs for patients (Jones, 2016). Elbow stiffness is defined as a loss of extension greater than 30°, being the most common outcome, and a flexion of less than 120°, which significantly affects arm functionality, since hardly compensated by other joints, with up to 12% of all elbow trauma acquiring a restriction in flexion that necessitate surgical approach (Akhtar et al., 2021; Murray et al., 2012; Myden and Hildebrand, 2011). The loss of motion after elbow injury can result from different intrinsic factors, such as intra-articular incongruity or adhesions, loose body osteophyte formation or malalignment of the articular surfaces, and extrinsic factors, as capsular collateral ligament and muscular contraction, as well as heterotopic ossification, although clinically it is common

* Corresponding author. Department of Territorial Care, Local Healthcare Unit, Piazza Sandro Pertini 10, 17100, Savona, SV, Italy.
E-mail address: a.piacenza@asl2.liguria.it (A. Piacenza).

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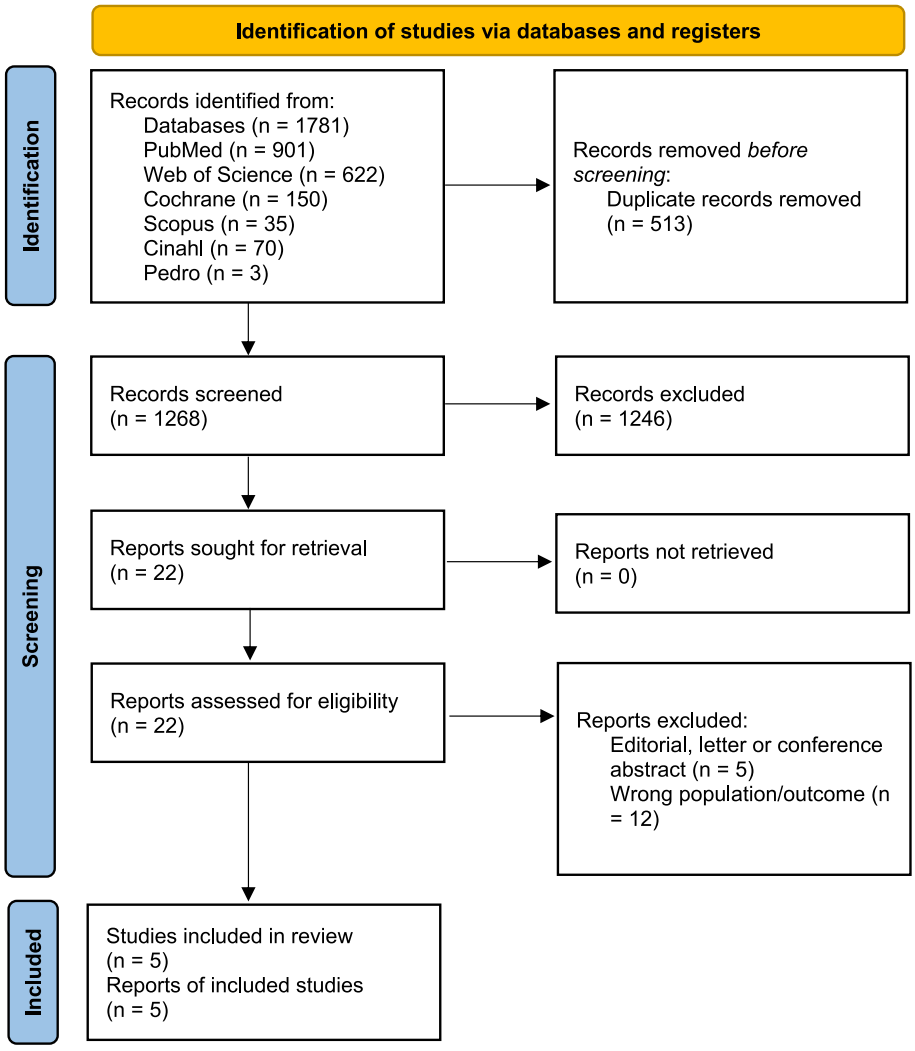


Fig. 1. PRISMA flow diagram.

to encounter a mixed condition, and joint motion recovery can be difficult and time-consuming (Akhtar et al., 2021).

In the post-traumatic elbow stiffness management, conservative treatment is recommended as the first-line approach, particularly within the first six months following the onset of stiffness (Siemensma et al., 2023; Yau et al., 2020). This is due to the considerable risks of post-operative complications associated with surgical release of a stiff elbow. However, despite the variety of approaches described for the conservative management of post-traumatic elbow stiffness, there appears to be limited evidence (Jones, 2016).

The purpose of this systematic review is to investigate and compare the effectiveness of conservative interventions in patients with elbow stiffness of traumatic or postoperative origin. For the purpose of this study, effectiveness was measured as the improvement in function, pain, and range of motion.

2. Methods

2.1. Study design

The protocol of this systematic review was pre-registered in the International Prospective Register of Systematic Reviews (PROSPERO: CRD42024517823) and followed the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) statement (Page et al., 2021) and Cochrane guidelines (Higgins et al., 2019).

2.2. Search strategy

The search strategy was conducted in PubMed (MEDLINE), Cochrane Library, CINHAL, Web of Science (WoS), Scopus, and Physiotherapy Evidence Database (PEDro) from inception to April 05, 2024. The following Medical Subject Headings were used in the search strategy: elbow, contracture, rehabilitation, conservative treatment, physical therapy modalities, and exercise. These terms were combined with other keywords and linked with the Boolean operators AND/OR. The search strategy used in each database is shown in Appendix A. A hand search of the reference lists of all the included studies was performed.

2.3. Screening and study eligibility criteria

The studies identified during the searches were screened to remove duplicates. Two reviewers (AP and AZ) independently assessed titles and abstracts of the publications retrieved during the electronic search based on the inclusion criteria. Differences during the screening process were discussed between the two reviewers until consensus was reached. If no consensus was obtained a third researcher (IV) was consulted to make a final decision.

The inclusion criteria were developed according to the Population, Intervention, Comparison, Outcome, and Study design (PICOS) framework. All the studies met the following criteria: Population: subjects suffering from post-traumatic elbow stiffness; Intervention: conservative

Table 1
Summary of the study information and results of the included studies.

Author and year (country)	Population and study design	Sample characteristics (male/female, age)	Interventions and follow-up (frequency)	Outcomes	RESULTS
Birinci et al. 2019 (Turkey)	Posttraumatic or post-operative elbow stiffness (RCT)	40 (M 16/F 24, mean age 41.34 ± 8.92)	Proprioceptive neuromuscular facilitation VS Static stretching (2x/week for 6 weeks) Follow-up: 6 week (discharge) and 10 week	DASH, active ROM, Pain, Kinesiophobia, QoL, Patient satisfaction	Significant difference at DASH($p = 0.03$) in favour of PNF group. Significant differences for elbow flexion AROM ($p = 0.04$), VAS-rest ($p = 0.03$) and VAS-activity ($p = 0.01$) in favour of PNF group. No between-groups differences in other outcomes.
Birinci et al. 2022 (Turkey)	Post-operative elbow stiffness (RCT)	50 (M 32/F 18, mean age 41.9 ± 10.86)	Graded Motor Imagery VS Structured exercise program (2x/week for 6 weeks) Follow-up: 6 week (discharge) and 12 week	DASH, Active ROM, Pain, Kinesiophobia, muscle strength of elbow flexors and extensors, grip strength, left-right discrimination, and patient satisfaction.	Significant difference for DASH score ($p > 0.01$) in favour of the GMI group. Significant differences for elbow flexion-extension AROM, VAS-activity, Kinesiophobia, recognition accuracy and response time of hand in the affected side, and muscle strength of elbow flexors and elbow extensors ($P < 0.05$) in favour of GMI group.
Lindenhovius et al. 2012 (USA)	Posttraumatic or post-operative elbow stiffness (RCT)	66 (M 40/F 26, mean age 44 ± 12)	Dynamic splints (6–8 h/day) VS Static progressive splints (30 min 3 times x day) Follow-up: 3, 6, and 12 months	DASH, ROM	No significant differences between groups
Bhosale et al. 2023 (India)	Post-operative elbow stiffness (RCT)	26 (M 16/F 10, mean age 41.07 ± 13.12)	Instrument-assisted soft tissue mobilization VS Muscle Energy Technique (2x/week for 3 weeks) Follow-up: 3 weeks (discharge)	DASH, Pain, ROM, Patient-specific functional scale	Significant differences at NPRS-activity, and Patient-specific functional scale in favour of IASTM group. No between-groups differences in other outcomes.
Faqih et al. 2019 (India)	Post-operative elbow stiffness (RCT)	30	MET immediately after immobilization removal VS MET after 1 week post removal of immobilization (6x/week for 3 weeks) Follow-up: 3 weeks (discharge)	DASH, Pain, ROM	Significant difference at DASH. No between-groups statistically significant differences in other outcomes.



Fig. 2. RoB 2.0 assessment.

approaches as physiotherapy interventions, splinting and bracing, exercise, and pharmacological approaches; Comparison: any conservative treatment; Outcomes: improvements in terms of function, pain, range of motion (ROM); Study design: randomized controlled trials. Studies were excluded if they: included subjects <18 years old; included patients with non-traumatic elbow injuries or post-burn elbow stiffness, neurological pathology causes increased muscle tone/spasticity, structural (bony) causes of elbow stiffness which may only respond to surgical intervention e.g., heterotrophic ossification, impinging osteophytes, or impinging hardware; the outcome variables reported were not the outcomes of interest or were not measured using a valid and reliable instrument. To increase the reliability of the review's findings, only randomized controlled trials (RCTs) were included (Evans, 2003).

2.4. Data extraction

Data were extracted using a standardized form that documents participants' characteristics (number, age, gender), selection criteria, interventions (modality, frequency, duration), control characteristics, outcome measures, follow-up periods, and results. Two independent reviewers (AP and AZ) extracted data and any discrepancy was resolved by an independent third reviewer (IV).

2.5. Risk of bias assessment

The risk of bias (RoB) assessment of the selected papers was performed by two independent assessors (AP, AZ) using the Revised Cochrane Risk of Bias tool (RoB2) (Sterne et al., 2019). This tool rates each paper according to potential bias arising from five main domains: randomization process, deviation from intended intervention, missing outcome data, measurement of the outcome and selection of the reported results. Judgments of each domain and overall risk of bias were based on an algorithm aimed to assign to each item a quality rating of low RoB, some concern or high RoB (Sterne et al., 2019). Disagreements were discussed between the two reviewers and any remaining disagreements were resolved through further discussion with a third researcher (IV).

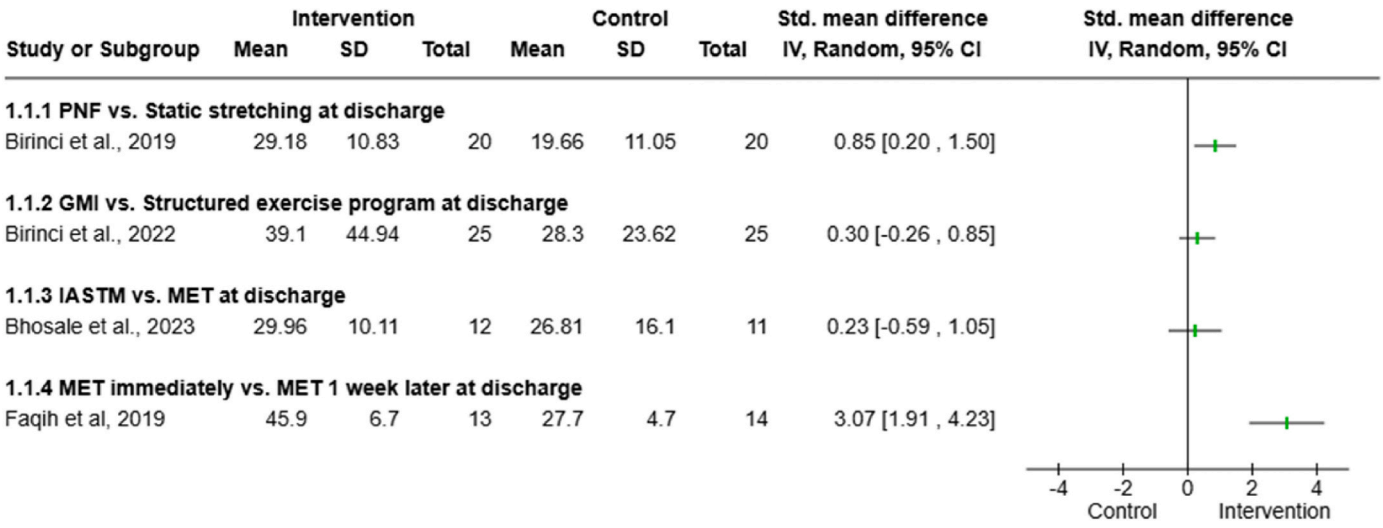


Fig. 3. – Elbow functionality at the end of the treatment. All analyses are described as standardized mean differences (SMD) with random effects. The effect sizes were calculated using mean change before-after the treatments. Higher values of mean for individual groups mean higher change (reduction of functional impairment).

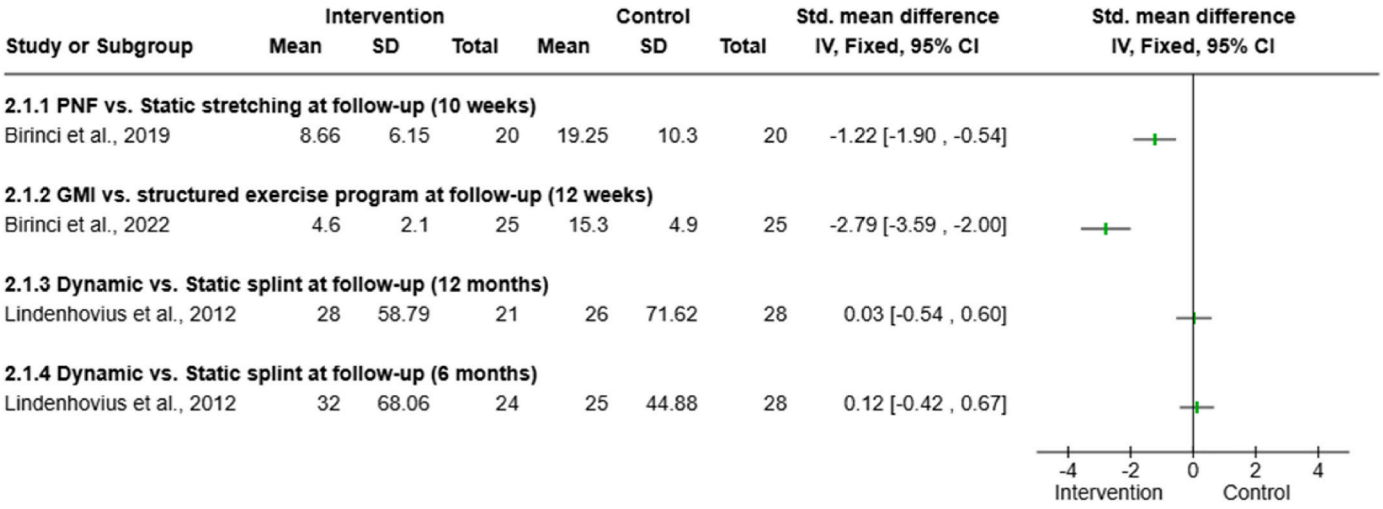


Fig. 4. – Elbow functionality at follow-up. All analyses are described as standardized mean differences (SMD) with random effects. The effect sizes were reported as mean DASH values at follow-up. Lower values for individual groups mean higher levels of functional status.

2.6. Data analysis and synthesis

Quantitative data collection was based on sample size, mean scores, and standard deviations. Between-group effect sizes were calculated where feasible. For ROM, pain, and function measures, which comprised solely continuous data, Standardized Mean Differences (SMD) (Hedge's g) were determined along with their 95% confidence intervals. Cohen's criteria (Cohen, 1988) were used to interpret effect sizes, meaning that 0.2, 0.5, and 0.8 were used to determine small, medium, and large effect sizes respectively. We visually assessed evidence of publication bias through funnel plots. Review Manager Software (RevMan 5.4) was used to perform the statistical analyses. Heterogeneity was explored based on clinical characteristics, RoB of the studies, and visually assessed in the forest plot. Since high heterogeneity of interventions and comparators did not allow for a metanalysis to be performed, a structured narrative synthesis was provided based on intervention type and follow-up duration.

2.7. Quality of evidence

The Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) framework was used by two reviewers (AP, AZ) and, in case of disagreements, a third author (IV) adjudicated to assess the confidence in the body of evidence reviewed (Guyatt et al., 2008). This framework suggests rating levels of results across five domains (risk of bias, inconsistency, indirectness, imprecision, and publication bias) as high, moderate, low or very low (Guyatt et al., 2011). For each outcome, evidence was downgraded in accordance of the presence of the following: risk of bias (downgraded by one level if the study was at some concerns of bias; and two levels if the study was at high risk of bias), inconsistency (evidence was downgraded by one level when results were from one single study), indirectness (low risk of indirectness was attributed by default because all outcomes included were deemed appropriate concerning the specificity of populations, interventions, comparators, and outcomes guaranteed by the eligibility criteria), imprecision (downgraded by one level if the estimates comes from small sample studies) (Guyatt et al., 2011), risk of publication bias (downgraded by one level if there was suspected publication bias).

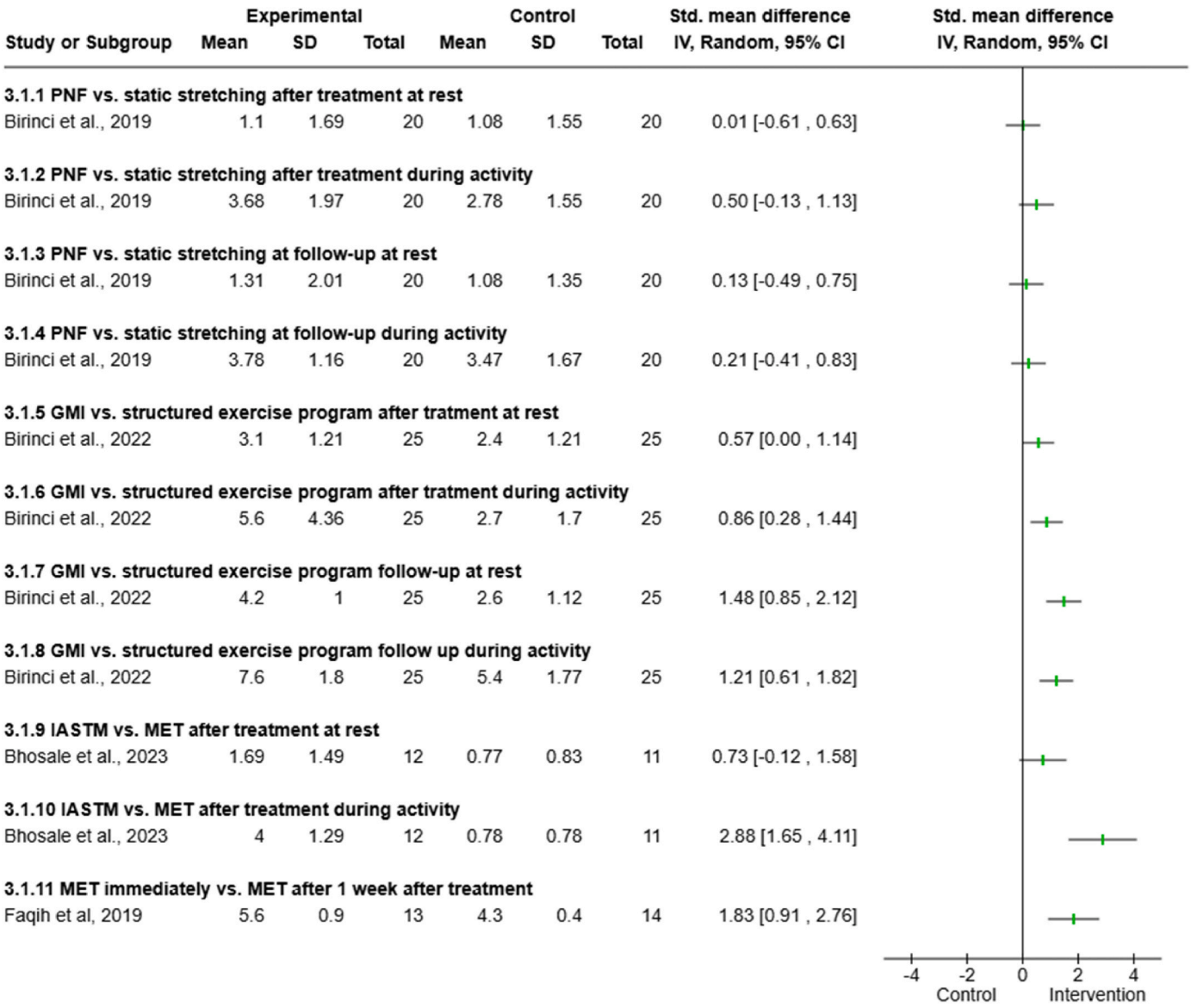


Fig. 5. – Pain intensity. All analyses are described as standardized mean differences (SMD) with random effects. The effect sizes were calculated using mean change before-after the treatments. Higher values of mean for individual groups mean higher change (reduction of pain).

3. Results

3.1. Study selection

The literature search produced 1781 records. Following removal of duplicates 1268 titles and abstracts were screened, 22 potential full texts were assessed using the eligibility criteria, and 5 original trials were included in the review (Fig. 1).

3.2. Studies characteristics

The 5 studies identified in the review analyzed different conservative approaches and included 212 patients. Table 1 shows study details. All participants included in the studies suffered from elbow stiffness as a consequence of a traumatic injury and/or after undergoing surgical intervention. Conservative interventions proposed in the studies encompass manual passive techniques as Muscle Energy Technique (MET), Proprioceptive Neuromuscular Facilitation (PNF), instrument-assisted soft tissue mobilization (IASTM), Graded Motor Imagery (GMI), and dynamic splint. Regarding comparators, conservative treatments included static stretching, MET, static splints and a structured exercise program composed of active and passive movements of the upper limb. All the studies assessed the elbow range of motion (ROM), and administered the Disability of the Arm, Shoulder and Hand (DASH) to investigate the functionality of the elbow, while in only 4 studies the pain domain was investigated by administering the VAS (visual analogue scale) or NPRS (numeric pain rating scale). Follow-up times vary between 3 weeks from the start of conservative treatment up to 12 months.

3.3. Risk of bias assessment

All the studies were evaluated using the RoB2 tool. Overall, three papers were rated as having low risk of bias, one as having some concerns of risk of bias, and one as having high risk of bias (Fig. 2). There was some concern about RoB in one study regarding randomization concealment of group allocation, and potential selection of reported results because the published protocol was not available (Faqih et al., 2019); while one study was rated as high risk of bias in the "deviations from the intended interventions" domain (Lindenhovius AL et al., 2012).

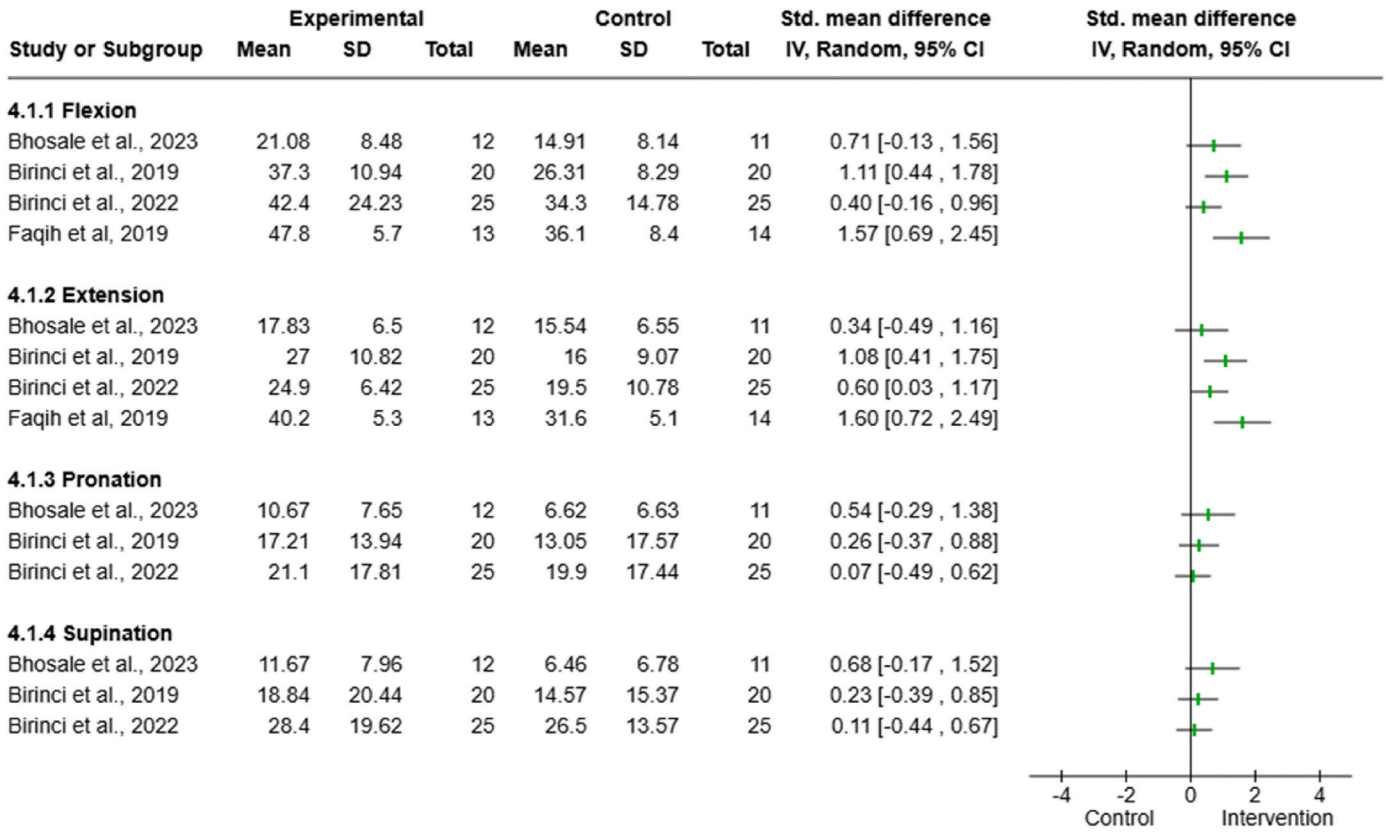


Fig. 6. – Range of motion at the end of the treatment. All analyses are described as standardized mean differences (SMD) with random effects. The effect sizes were calculated using mean change before-after the treatments. Higher values of mean for individual groups mean higher change (improvement of range of motion).

3.4. Outcomes

3.4.1. Elbow functionality

Elbow functionality was evaluated in all the studies examined by administering the DASH questionnaire. All studies provided quantitative data; however, due to high heterogeneity of interventions and comparators, and different follow-up timing, it was not possible to perform a meta-analysis. Effect sizes were calculated for each study and presented in Figs. 3–4, and GRADE analysis was presented in Fig. 7. In the short term, one study (Faqih et al., 2019) showed higher improvement at the end of the treatment in the group where MET was applied immediately post immobilization compared to the group where MET was started one week after post removal of immobilization (SMD = 3.07; 95%CI [1.91, 4.23]), but the quality of evidence was very low (downgraded for risk of bias, inconsistency and imprecision). Furthermore, large effect sizes were shown for PNF versus static stretching (Birinci et al., 2019) at the end of the treatment (SMD = 0.85; 95%CI [0.20, 1.50]), and after six weeks from discharge (SMD = −1.22; 95%CI [−1.90, −0.54]), but the quality of evidence was low (downgraded for inconsistency and imprecision). In the long term, large effect size was observed in favour of GMI when compared to a structured exercise program (Birinci et al., 2022) involving active and passive movements of the upper limb (SMD = −2.79; 95%CI [−3.59, −2.00]), although the quality of evidence was low (downgraded for inconsistency and imprecision). No differences were noted for other interventions in the short and long term.

3.4.2. Pain intensity

Four studies (Bhosale and Kolke Pt, 2023; Birinci et al., 2019, 2022; Faqih et al., 2019) investigated the effectiveness of different conservative approaches on diminishing pain intensity. Effect sizes were calculated and represented in Fig. 5 while GRADE analysis was presented in Fig. 8. Moderate to large effect size was shown for GMI versus control (Birinci et al., 2022) for both pain at rest (SMD = 0.57; 95%CI [0.00, 1.14]), and pain during activity (SMD = 0.86; 95%CI [0.28, 1.44]) at the end of the treatment, and after 12 weeks from enrolment (SMD = 1.48; 95%CI [0.85, 2.12] at rest, SMD = 1.21; 95%CI [0.61, 1.82] during activity), although the quality of evidence was low (downgraded for inconsistency and imprecision). Moreover, moderate to large effect sizes were observed for PNF (SMD = 0.50; 95%CI [−0.13, 1.13]) (Birinci et al., 2019), and MET (SMD = 1.83; 95%CI [0.91, 2.76]) (Faqih et al., 2019) in comparison to control groups at the end of the treatment, demonstrating a reduction in pain during activity and rest, respectively. However, the quality of the evidence was rated as low to very low. For the other comparisons against control, no differences were found.

3.4.3. Range of motion

Four randomized controlled trials reported the efficacy of conservative treatments at discharge on flexion-extension ROM for post-traumatic elbow stiffness (Bhosale and Kolke Pt, 2023; Birinci et al., 2019, 2022; Faqih et al., 2019), while only three studies investigated the efficacy on pronation-supination ROM (Bhosale and Kolke Pt, 2023; Birinci et al., 2019, 2022). Effect sizes were calculated for each study and presented in Fig. 6. The subgroup analyses indicated that MET could significantly improve flexion-extension ROM when conservative treatment was started immediately after the immobilization period (Faqih et al., 2019), but no differences were found when compared to IASTM (Bhosale and Kolke Pt, 2023). Moreover, moderate and large effect size were shown in favour of PNF (Birinci et al., 2019) and GMI (Birinci et al., 2022) to improve flexion-extension ROM. When investigating the efficacy of conservative treatments on pronation-supination ROM, no differences were found among the proposed conservative treatments.

Certainty assessment							№ of patients		Effect	Certainty
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Treatment of interest	Comparator of interest	Absolute (95% CI)	
Elbow Functionality – PNF vs. static stretching at the end of the treatment (6 weeks), assessed with DASH (Scale from: 0 to 100)										
1(Birinci et al., 2019)	randomised trials	not serious	NA	not serious	serious	none	20	20	SMD **0.85** SD higher (0.20 higher to 1.50 higher)	⊕⊕◯◯[a] Low
Elbow Functionality – GMI vs. structured exercise program at the end of the treatment (6 weeks), assessed with DASH (Scale from: 0 to 100)										
1(Birinci et al., 2022)	randomised trials	not serious	NA	not serious	serious	none	25	25	SMD **0.30** SD higher (0.26 lower to 1.62 lower)	⊕⊕◯◯[a] Low
Elbow Functionality – IASTM vs. MET at the end of the treatment (3 weeks), assessed with DASH (Scale from: 0 to 100)										
1(Bhosale & Kolke, 2023)	randomised trials	not serious	NA	not serious	serious	none	12	11	SMD **0.23** SD higher (0.59 lower to 1.05 higher)	⊕⊕◯◯[a] Low
Elbow Functionality – MET immediately vs. MET after one week at the end of the treatment (3 weeks), assessed with DASH (Scale from: 0 to 100)										
1(Faqih et al., 2019)	randomised trials	serious	NA	not serious	serious	none	13	14	SMD **3.07** SD higher (1.91 higher to 4.23 higher)	⊕◯◯◯[b] Very low
Elbow Functionality – PNF vs. static stretching at follow-up (10 weeks), assessed with DASH (Scale from: 0 to 100)										
1(Birinci et al., 2019)	randomised trials	not serious	NA	not serious	serious	none	20	20	SMD **1.22** SD lower (1.9 lower to 0.54 lower)	⊕⊕◯◯[a] Low
Elbow Function – GMI vs. structured exercise program at follow-up (12 weeks), assessed with DASH (Scale from: 0 to 100)										
1(Birinci et al., 2022)	randomised trials	not serious	NA	not serious	serious	none	25	25	SMD **2.79** SD lower (3.59 lower to 2 lower)	⊕⊕◯◯[a] Low
Elbow Function – Dynamic splinting vs. static splinting at follow-up (6 months), assessed with DASH (Scale from: 0 to 100)										
1(Lindenhovius AL et al., 2012)	randomised trials	very serious	NA	not serious	serious	none	24	28	SMD **0.12** SD higher (0.42 lower to 0.67 higher)	⊕◯◯◯[c] Very low
Elbow Function – Dynamic splinting vs. static splinting at follow-up (12 months), assessed with DASH (Scale from: 0 to 100)										
1(Lindenhovius AL et al., 2012)	randomised trials	very serious	NA	not serious	serious	none	21	28	SMD **0.03** SD higher (0.42 lower to 0.67 higher)	⊕◯◯◯[c] Very low

Fig. 7. GRADE ratings for Elbow functionality by comparison.
CI: Confidence interval; SMD: Standardized Mean difference; SD: Standard Deviation.
Explanations.
a. Downgraded one level for inconsistency (single study) and one level for imprecision (small sample size).
b. Downgraded one level for risk of bias (study classified as some concerns of bias), one level for inconsistency (single study) and one level for imprecision (small sample size).
c. Downgraded two levels for risk of bias (study classified as high risk of bias), one level for inconsistency (single study) and one level for imprecision (small sample size).
High: We are very confident that the true effect is close to the estimate of the effect.
Moderate: We are moderately confidence in the effect estimate. The true effect is close to the estimate of the effect, but the result can be different. Low: Confidence in the effect estimate is limited, the true effect can be substantially different from the estimate of the effect.
Very Low: There is little confidence in the effect estimate, the true effect is likely to be substantially different from the estimate effect.

4. Discussion

This systematic review aimed to summarize the available evidence for the effectiveness of conservative interventions in patients with elbow stiffness of traumatic or postoperative origin, in terms of improvement in function, pain, and ROM. Results indicate that conservative treatments suggested in the literature showed substantial differences in terms of techniques, dosage, timing and procedures applied, making them difficult to compare. Indeed, proposed conservative interventions range from soft tissue to splinting techniques, to instrumented assisted manual therapy or approaches more focused on activating the premotor and primary motor cortex via motor-cognitive strategies. Overall, results indicate that initiating conservative treatment early leads to improved outcomes in function, pain reduction, and ROM in the short term. Although the literature suggests that the majority of patients are likely to regain functional range of motion within a year following elbow trauma, it is widely agreed that prompt initiation of the rehabilitation process, whether active or passive, is a safe and effective method for restoring elbow function and facilitating a quick return to work activities, without increasing the likelihood of complications (Giannicola et al., 2014; Harding et al., 2011; Iordens et al., 2017; Jones, 2016; Siemensma et al., 2023). Despite the etiopathogenesis of post-traumatic elbow stiffness is not fully understood, the literature indicates that an increase in the extracellular matrix occurs soon after trauma or surgery, leading to a subsequent contracture (Doornberg et al., 2014). Therefore, conservative strategies primarily ought to focus on treating local structures to impede this remodeling process and prevent the onset of elbow stiffness. It is noteworthy that the considered RCTs did not detail the evaluation

Certainty assessment							№ of patients		Effect	Certainty
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Treatment of interest	Comparator of interest	Absolute (95% CI)	
Pain intensity – PNF vs. static stretching at the end of the treatment (6 weeks) at rest, assessed with VAS (Scale from: 0 to 10)										
1(Birinci et al., 2019)	randomised trials	not serious	NA	not serious	serious	none	20	20	SMD **0.01** SD higher (0.61 lower to 0.63 higher)	⊕⊕◯◯[a] Low
Pain intensity – PNF vs. static stretching at the end of the treatment (6 weeks) during activity, assessed with VAS (Scale from: 0 to 10)										
1(Birinci et al., 2019)	randomised trials	not serious	NA	not serious	serious	none	20	20	SMD **0.50** SD higher (0.13 lower to 1.13 higher)	⊕⊕◯◯[a] Low
Pain intensity – PNF vs. static stretching at follow-up (10 weeks) at rest, assessed with VAS (Scale from: 0 to 10)										
1(Birinci et al., 2019)	randomised trials	not serious	NA	not serious	serious	none	20	20	SMD **0.13** SD higher (0.49 lower to 0.75 higher)	⊕⊕◯◯[a] Low
Pain intensity – PNF vs. static stretching at follow-up (10 weeks) during activity, assessed with VAS (Scale from: 0 to 10)										
1(Birinci et al., 2019)	randomised trials	not serious	NA	not serious	serious	none	20	20	SMD **0.21** SD higher (0.41 lower to 0.83 higher)	⊕⊕◯◯[a] Low
Pain intensity – GMI vs. structured exercise program at the end of the treatment (6 weeks) at rest, assessed with VAS (Scale from: 0 to 10)										
1(Birinci et al., 2022)	randomised trials	not serious	NA	not serious	serious	none	25	25	SMD **0.57** SD higher (0 to 1.14 higher)	⊕⊕◯◯[a] Low
Pain intensity – GMI vs. structured exercise program at the end of the treatment (6 weeks) during activity, assessed with VAS (Scale from: 0 to 10)										
1(Birinci et al., 2022)	randomised trials	not serious	NA	not serious	serious	none	25	25	SMD **0.86** SD higher (0.28 higher to 1.44 higher)	⊕⊕◯◯[a] Low
Pain intensity – GMI vs. structured exercise program at follow-up (12 weeks) at rest, assessed with VAS (Scale from: 0 to 10)										
1(Birinci et al., 2022)	randomised trials	not serious	NA	not serious	serious	none	25	25	SMD **1.48** SD higher (0.85 higher to 2.12 higher)	⊕⊕◯◯[a] Low
Pain intensity – GMI vs. structured exercise program at follow-up (12 weeks) during activity, assessed with VAS (Scale from: 0 to 10)										
1(Birinci et al., 2022)	randomised trials	not serious	NA	not serious	serious	none	25	25	SMD **1.21** SD higher (0.61 higher to 1.82 higher)	⊕⊕◯◯[a] Low
Pain intensity – IASTM vs. MET at the end of the treatment (3 weeks) at rest, assessed with NPRS (Scale from: 0 to 10)										
1(Bhosale & Kolke, 2023)	randomised trials	not serious	NA	not serious	serious	none	12	11	SMD **0.73** SD higher (0.12 lower to 1.58 higher)	⊕⊕◯◯[a] Low
Pain intensity – IASTM vs. MET at the end of the treatment (3 weeks) during activity, assessed with NPRS (Scale from: 0 to 10)										
1(Bhosale & Kolke, 2023)	randomised trials	not serious	NA	not serious	serious	none	12	11	SMD **2.88** SD higher (1.65 higher to 4.11 higher)	⊕⊕◯◯[a] Low
Pain intensity – MET immediately vs. MET after one week at the end of the treatment (3 weeks), assessed with VAS (Scale from: 0 to 10)										
1(Faqih et al., 2019)	randomised trials	serious	NA	not serious	serious	none	13	14	SMD **1.83** SD higher (0.91 higher to 2.76 higher)	⊕◯◯◯[b] Very low

Fig. 8. GRADE ratings for pain intensity by comparison.
CI: Confidence interval; SMD: Standardized Mean difference; SD: Standard Deviation.
Explanations.
a. Downgraded one level for inconsistency (single study) and one level for imprecision (small sample size).
b. Downgraded one level for risk of bias (study classified as some concerns of bias), one level for inconsistency (single study) and one level for imprecision (small sample size).
High: We are very confident that the true effect is close to the estimate of the effect.
Moderate: We are moderately confidence in the effect estimate. The true effect is close to the estimate of the effect, but the result can be different. Low: Confidence in the effect estimate is limited, the true effect can be substantially different from the estimate of the effect.
Very Low: There is little confidence in the effect estimate, the true effect is likely to be substantially different from the estimate effect.

process used to determine the preference for one type of treatment over another. Indeed, a careful assessment of the most restricted movement directions can aid in identifying the structures that are most implicated, such as the humeroulnar, radioulnar, or proximal radioulnar joints, as well as musculotendinous or capsuloligamentous structures. This could guide physiotherapists in focusing on the most limited movement directions and the most efficacious treatment approaches to impact the implicated structures more effectively. Moreover, a careful analysis of

movement directions can more accurately identify limitations not directly associated with local tissue dysfunctions, but may instead be more linked to phenomena such as pain guarding, protective responses or kinesiophobia. Enhancing treatments with motor-cognitive strategies, rather than solely relying on passive approaches of local structures, has proven to be an effective way to alleviate pain and mitigate the adverse effects associated with immobilization in the short to medium term. While the literature lacks a systematic review of conservative treatment, three reviews assessed the efficacy of splints (Chen et al., 2017; Müller et al., 2013; Veltman et al., 2015). Findings from these studies suggest that both dynamic orthoses and static progressive splints effectively are effective in treating elbow stiffness, regardless of the etiology. In our review, no differences were found among dynamic and static splints. Nevertheless, caution is advised due to the evidence being of very low quality, as evaluated by the GRADE approach.

The analysis of the studies has revealed that only two papers have investigated patient satisfaction following conservative treatment (Birinci et al., 2019, 2022). Recent studies indicate that patient satisfaction is not solely dependent on treatment outcomes (Hush et al., 2011; Rossettini et al., 2020). Therefore, while enhancing clinical results is beneficial, it is not the only factor influencing patient satisfaction. An important factor that needs to be highlighted is that the clinical efficacy of therapeutic interventions is reliant on a complex interplay of multiple factors. It highlights that the outcomes of each therapeutic intervention are contingent upon the interplay among various factors, delineating a multidimensional construct shaped by individual patient/provider characteristics, as well as clinical and contextual factors (Liu et al., 2022; Rossettini et al., 2020).

Although this review did not examine the effectiveness of conservative treatment in comparison to invasive interventions, it is important to recognize that the body of evidence which supports conservative treatment is less reliable than an invasive approach. Given the higher risk of complications associated with more invasive procedures, conservative management should be the standard first-line approach, due to the lower risk of complications (Araghi et al., 2010; Cai et al., 2015, 2016; Cui et al., 2019; Iordens et al., 2017; Zheng et al., 2018). Nonetheless, recent research indicates that botulinum toxin injections during surgery may enhance functional recovery without increasing complication rates (Bäcker et al., 2021; Freibott et al., 2020). Consequently, future research should prioritize comparing conservative and invasive treatment outcomes, including patient satisfaction. Additionally, it is imperative to establish a standardized treatment protocol with defined follow-up durations and outcome metrics to facilitate effective comparisons and determine the most efficacious conservative treatment.

4.1. Strengths and limitations

To our knowledge, this SR constitutes the first attempt to investigate and compare the effectiveness of conservative treatments in patients with elbow stiffness of traumatic or postoperative origin. We adhered to a predefined protocol that was published in the PROSPERO register (CRD42024517823), and conducted the review in adherence with the most accepted scientific standards. However, despite employing a thorough methodology, the goals of this systematic review were not entirely achieved. This was due to the limited number of studies analyzed, which showed a heterogeneous spectrum of conservative approaches, the small sample size of participants, short follow-up measurements, and the evidence being of low to very low quality. Clinically, another significant concern is that while many studies focus on short-term results, only a select few examine the medium and long-term outcomes, thus limiting their interpretation in terms of applicability and reproducibility. Moreover, a limitation of this paper may be seen in the exclusion of studies comparing the effectiveness of conservative treatments to surgical or invasive approaches, as well as those examining the impact of physical therapy after surgical procedures for post-traumatic elbow stiffness.

5. Conclusion

Overall, the available evidence on the topic is limited. Despite conservative management should be adopted as the first choice of treatment in the management of post-traumatic elbow stiffness, the most effective conservative treatment in reducing disability, pain, and restriction of mobility could not be determined. Hence, it is advisable to carry out further high-quality research to explore which conservative methods, types, and dosages are most effective.

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Registration

PROSPERO - CRD42024517823 - Effectiveness of conservative treatment in the management of post-traumatic elbow stiffness.

Declaration competing of interest

None.

CRediT authorship contribution statement

Alberto Piacenza: Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Andrea Zerilli:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Ilenia Viccari:** Writing – review & editing, Supervision, Investigation, Formal analysis, Data curation. **Greta Castelli:** Project administration, Investigation, Formal analysis, Data curation, Conceptualization.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.msksp.2024.103194.

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